

Qatar-1b

R-1971

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130520 · created 2026-08-06T23:58:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456431.486465 +/- 4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.058
Transit depth (Rp/Rs)^2	3.69 +/- 2.24 [%]
Orbital Inclination (inc)	85.6 +/- 2.79
Airmass coefficient 1 (a1)	3.17 +/- 0.37
Airmass coefficient 2 (a2)	-0.82 +/- 0.29
Scatter in the residuals of the lightcurve fit is	12.01 %
Best Comparison Star	#3 - [57, 160]
Optimal Aperture	14.88
Optimal Annulus	54.73
Transit Duration (day)	0.0295 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.058 vs 0.1463 (deviation 0.63 σ)
Duration fit vs published	0.0295 d vs 0.0692 d (deviation 4.44 σ)
Mid-time O-C	-2044.9 min
Depth SNR	2.6
Residual scatter	12.01 %

Light curve

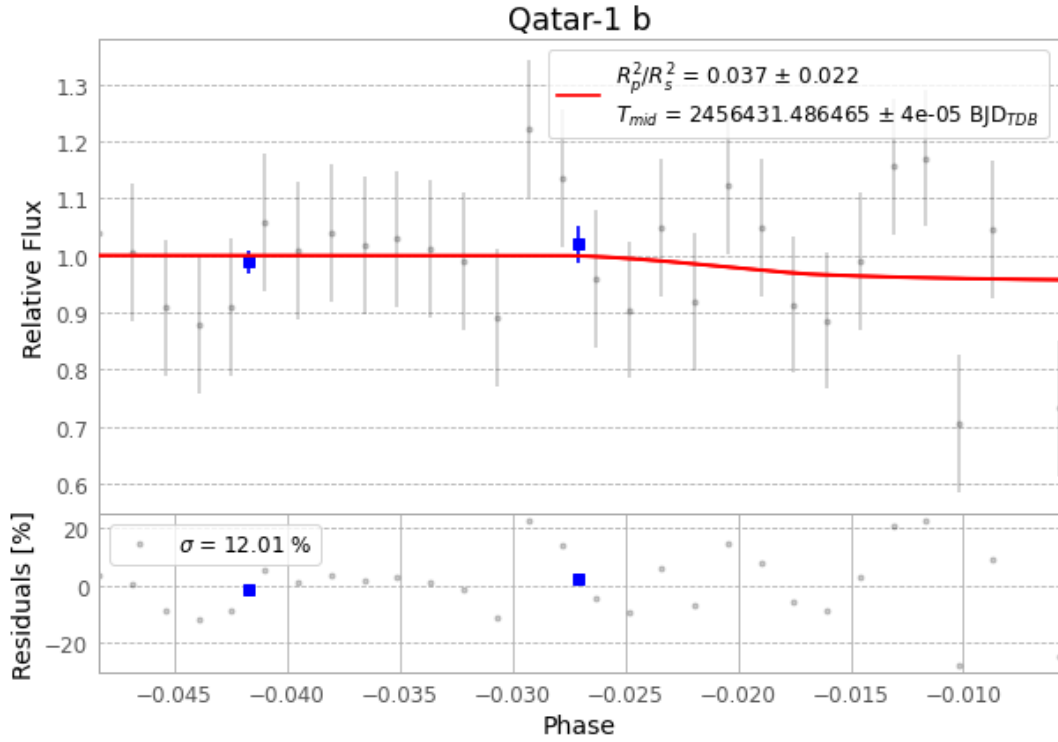


Figure 1 — FinalLightCurve_Qatar-1 b_2013-05-20.png (SHA-256 966f58219495012b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 449], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a682f03dc176cf3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), Qatar-1130520080612.FITS → Qatar-1130520112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130520.log (059815d87970...), FinalLightCurve_Qatar-1 b_2013-05-20.png (966f58219495...), FinalLightCurve_Qatar-1 b_2013-05-20.csv (d56acefc2454...)

Compute resources

Wall time 265.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:58Z.